OFFICIAL EDUCATOR DOSSIER & INNOVATION PORTFOLIO

Holder: Tshingombe Tshitadi Fiston | Specialization: Electrical Engineering, Pedagogical Innovation & Didactic Systems

1. Professional Credentials & Licenses

- **Secondary Technical Diploma:** ITI Kitomesa
- **Pedagogical Engineering:** ISPT Kinshasa (*Diplôme d'Ingénieur Pédagogique*)
- **Vocational Certification:** INPP (*Institut National de Préparation Professionnelle*)
- **Quality Assurance:** SAQA Assessment Awareness Accreditation
- **AI Research Competency:** Advanced AI-Assisted Technical Documentation Certificate

2. Patent & Brevet d'Invention

- **Title:** Modular Didactic Smart Grid Fault Coordination Simulator
- **Registration Ref:** *BREVET/INNO-417-EDUT*
- **Core Discovery:** Integrated hardware-software learning panel for $V_{L\text{-}L} = 380V/220V$ power systems and ANSI 25/51 relay coordination.
- **Application:** Vocational training, lab workshops, and automated compliance testing.

3. Didactic Panel & Laboratory Architecture (DL SGWD System)

The innovation centers around an integrated engineering lab bench combining alternative energy sources (wind, hydro, solar PV), fault protection modules, power factor control, and SCADA monitoring. Designed for multi-tier technical training.

Subsystem Module	Technical Specification	Didactic Objective	Integration Status
Main Power Supply (DL 10051)	3-phase $380V / 220V$, overload protection	Power distribution safety and grid isolation fundamentals	Active
Renewable Sources (DL 10060-64)	Wind turbine, Hydro turbine, PV solar array	$I\text{-}V$ curve tracing and MPPT optimization analysis	Active
Fault Protection (DL 10071)	ANSI 25 (Synchronization) & ANSI 51 (Overcurrent)	Fault isolation, protection coordination, and THD analysis	Active
SCADA Supervision (DL SGWD)	MODBUS RTU, Ethernet TCP/IP interface	Supervisory control, data logging, and automated telemetry	Active

4. Skill Matrix & Vocational Levels

Skill Domain	Proficiency	Equivalent VQA
Power Systems Analysis	84%	Level 4 (Advanced)
Renewable Energy Tech	76%	Level 3 (Senior)
SCADA & MODBUS RTU	66%	Level 3 (Senior)
Lab Development & Didactics	95%	Level 5 (Expert)
Curriculum & VBA Automation	96%	Level 5 (Expert)

5. Training Phase & Lab Workshop Structure

- **Phase Foundation (Beginner):** Elementary engineering principles, physics apparatus, mathematics foundations, and electrical components.
- **Phase Intermediate (Junior):** Circuit analysis labs, bench design, and modular workshop execution.
- **Phase Senior (Cadet):** Power system stability, moderator-designed assessments, and compliance auditing.
- **Tableur & VBA Tools:** Automated management spreadsheets for curriculum logging, test trade tracking, and repository sync.

6. Summary Table of Course Modules & Assessment Framework

Curriculum Module	Phase Level	Learning Hours	ECTS Credits	Assessment Mode	Moderation Status
MOD-BEG-01: Load Analysis & Basic Circuits	Beginner (Elementary)	40 hrs	4 ECTS	Practical Workshop & Written	Moderated
MOD-INT-02: Renewable Microgrid Integration	Intermediate (Junior)	30 hrs	5 ECTS	Simulator Lab Test	Moderated
MOD-SEN-03: Fault Protection & SCADA Ops	Senior (Cadet)	28 hrs	5 ECTS	Trade Certification Exam	Moderated
MOD-ADV-04: Smart Grid Capstone Project	Advanced (Senior)	10 hrs	6 ECTS	Peer Review & Panel Defense	Pending